
CHAPTER 3

NAIVE BAYES CLASSIFICATION

Naive Bayes Classification is a probabilistic model used in supervised learning, where it learns from labeled training data to classify new, unseen data points. It applies Bayes' Theorem with the naive assumption that all input features are conditionally independent given the class label.

Despite its simplicity, it performs surprisingly well in many real-world tasks—especially in text classification problems like spam detection, sentiment analysis, and document categorization.

3.1 Key Idea

The model computes the **posterior probability** of each class given the input features:

$$P(C_k \mid x_1, x_2, \dots, x_n) = \frac{P(C_k) \cdot P(x_1, x_2, \dots, x_n \mid C_k)}{P(x_1, x_2, \dots, x_n)} \quad (3.1)$$

Using the **conditional independence assumption**:

$$P(x_1, x_2, \dots, x_n \mid C_k) = \prod_{i=1}^n P(x_i \mid C_k) \quad (3.2)$$

The posterior simplifies to:

$$P(C_k \mid x_1, x_2, \dots, x_n) \propto P(C_k) \cdot \prod_{i=1}^n P(x_i \mid C_k) \quad (3.3)$$

The predicted class is the one with the highest posterior probability.

3.2 Example: Why Independence Helps?

Problem: Classify emails as spam or not spam based on three features:

- x_1 : Presence of the word "free".

- x_2 : Presence of the word "win".
- x_3 : Presence of the word "money".

Without Independence:

- Joint probability $P(x_1, x_2, x_3|\text{Spam})$ requires estimating $2^3 = 8$ probabilities.
- For n features, 2^n combinations need estimation.

With Independence:

$$P(x_1, x_2, x_3|\text{Spam}) = P(x_1|\text{Spam}) \cdot P(x_2|\text{Spam}) \cdot P(x_3|\text{Spam}).$$

- Only 3 probabilities, $P(x_1|\text{Spam})$, $P(x_2|\text{Spam})$, $P(x_3|\text{Spam})$ need estimation.

3.3 Steps in Naive Bayes Classification

1. **Prepare Labeled Data:** Each data point has features $(x_1, x_2, \dots, x_n)$ and a class label.
2. **Estimate Probabilities:** From training data, calculate:
 - Prior probabilities $P(C_k)$
 - Likelihoods $P(x_i | C_k)$
3. **Apply Bayes' Rule:** Use the formula to compute posterior probabilities.
4. **Predict Class:** Choose the class with the highest posterior.

3.4 Example: Probability of HIV

Problem: Calculate the probability of having HIV after a positive test result.

Given:

- HIV prevalence, $P(HIV) = 0.008$
- Test sensitivity, $P(T|HIV) = 0.95$
- Test specificity, $P(\neg T|\neg HIV) = 0.95$

Solution:

$$P(HIV|T) \propto P(T|HIV) \cdot P(HIV) = (0.95) \cdot (0.008) = 0.0076$$

$$P(\neg HIV|T) \propto P(T|\neg HIV) \cdot P(\neg HIV) = (0.05) \cdot (0.992) = 0.0496$$

Conclusion: Even with a positive result, the probability of having HIV is low due to the low prior probability.

3.5 Example: Naive Bayes for Spam Detection

Problem: Classify an email as "Spam" or "Not Spam" based on the occurrence of the words "Free", "Money" and "Win."

Dataset: The dataset consists of 50 spam emails and 50 non-spam emails. The count of the presence of words "Free", "Win", and "Money" is given in the dataset.

Word	Spam	Non-spam
"Free"	30	5
"Win"	25	10
"Money"	20	5

Table 3.1: Keyword Frequencies in Emails

Prior Probabilities:

$$P(S) = \frac{\text{Total Spam}}{\text{Total Emails}} = \frac{50}{100} = 0.5$$

$$P(\neg S) = \frac{\text{Total Non-spam}}{\text{Total Emails}} = \frac{50}{100} = 0.5$$

Likelihood Probabilities:

$$P(\text{Free}|S) = \frac{\text{Spam emails with "Free"}}{\text{Total Spam}} = \frac{30}{50} = 0.6$$

$$P(\text{Win}|S) = \frac{\text{Spam emails with "Win"}}{\text{Total Spam}} = \frac{25}{50} = 0.5$$

$$P(\text{Money}|S) = \frac{\text{Spam emails with "Money"}}{\text{Total Spam}} = \frac{20}{50} = 0.4$$

$$P(\text{Free}|\neg S) = \frac{\text{Non-spam emails with "Free"}}{\text{Total Not Spam}} = \frac{5}{50} = 0.1$$

$$P(\text{Win}|\neg S) = \frac{\text{Non-spam emails with "Win"}}{\text{Total Non-spam}} = \frac{10}{50} = 0.2$$

$$P(\text{Money}|\neg S) = \frac{\text{Non-spam emails with "Money"}}{\text{Total Non-spam}} = \frac{5}{50} = 0.1$$

Posterior Calculation:

$$\begin{aligned}
 P(\text{Spam}|\text{Free, Win, Money}) &\propto P(\text{Free}|S) \cdot P(\text{Win}|S) \cdot P(\text{Money}|S) \cdot P(S) \\
 &= 0.6 \cdot 0.5 \cdot 0.4 \cdot 0.5 \\
 &= 0.06
 \end{aligned}$$

$$\begin{aligned}
 P(\neg S | \text{Free, Win, Money}) &\propto P(\text{Free} | \neg S) \cdot P(\text{Win} | \neg S) \cdot P(\text{Money} | \neg S) \cdot P(\neg S) \\
 &= 0.1 \cdot 0.2 \cdot 0.1 \cdot 0.5 \\
 &= 0.001
 \end{aligned}$$

Comparing the results, we can say that there is higher probability that the email is spam if all three words "Free", "Win" and "Money" are present in the email.

3.6 Example: Predicting Tennis Play from dataset

Dataset:

Sl. No.	Outlook	Temp.	Humidity	Windy	Play Tennis?
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

Table 3.2: Weather Dataset for Tennis Prediction

Computing Joint Probabilities

Predict whether to play tennis given: Outlook = Sunny, Temperature = Hot, Humidity = High, Windy = False

Learning Phase: Computing Probabilities

Step 1: Compute Class Priors

Class	Count	Prior Probability
Yes	9	$P(Y) = \frac{9}{14} = 0.64$
No	5	$P(N) = \frac{5}{14} = 0.35$

Table 3.3: Class Priors for the Tennis Dataset

Step 2: Compute Conditional Probabilities

For each feature and class, calculate $P(\text{Feature}|\text{Class})$.

Conditional Probabilities

Outlook

Outlook	Class: Yes	Class: No	Total
Sunny	$\frac{2}{9}$	$\frac{3}{5}$	5
Overcast	$\frac{4}{9}$	$\frac{0}{5}$	4
Rainy	$\frac{3}{9}$	$\frac{2}{5}$	5

Table 3.4: Conditional Probabilities for Outlook

Temperature

Temperature	Class: Yes	Class: No	Total
Hot	$\frac{2}{9}$	$\frac{2}{5}$	4
Mild	$\frac{4}{9}$	$\frac{2}{5}$	6
Cool	$\frac{3}{9}$	$\frac{1}{5}$	4

Table 3.5: Conditional Probabilities for Temperature

Humidity

Humidity	Class: Yes	Class: No	Total
High	$\frac{3}{9}$	$\frac{4}{5}$	7
Normal	$\frac{6}{9}$	$\frac{1}{5}$	7

Table 3.6: Conditional Probabilities for Humidity

Windy

Windy	Class: Yes	Class: No	Total
False	$\frac{6}{9}$	$\frac{2}{5}$	8
True	$\frac{3}{9}$	$\frac{3}{5}$	6

Table 3.7: Conditional Probabilities for Windy

Summary of Learned Probabilities

Feature	$P(\text{Feature} \text{Y})$	$P(\text{Feature} \text{N})$
Outlook = Sunny	0.22	0.6
Temperature = Hot	0.22	0.4
Humidity = High	0.33	0.8
Windy = False	0.66	0.4

Table 3.8: Likelihoods for Tennis Prediction

Final Decision

We compute the posterior probabilities using Bayes' Rule:

$$\begin{aligned}
 P(\text{Y}|\text{X}) &\propto P(\text{Y}) \cdot P(\text{S}|\text{Y}) \cdot P(\text{Hot}|\text{Y}) \cdot P(\text{High}|\text{Y}) \cdot P(\text{False}|\text{Y}) \\
 &= (0.64)(0.22)(0.22)(0.33)(0.66) = 0.007
 \end{aligned}$$

$$\begin{aligned}
 P(\text{N}|\text{X}) &\propto P(\text{N}) \cdot P(\text{S}|\text{N}) \cdot P(\text{Hot}|\text{N}) \cdot P(\text{High}|\text{N}) \cdot P(\text{False}|\text{N}) \\
 &= (0.36)(0.6)(0.4)(0.8)(0.4) = 0.046
 \end{aligned}$$

Result: The model predicts that we should **not** play tennis.

3.7 Example: Bayesian Diagnosis with another HIV Example

Single Positive Test

Given:

- $P(\text{HIV}) = 0.008$, $P(\text{Positive}|\text{HIV}) = 0.95$
- $P(\text{Positive}|\neg\text{HIV}) = 0.05$, $P(\neg\text{HIV}) = 0.992$

Using Bayes' theorem:

$$P(\text{HIV}|\text{Positive}) = \frac{0.95 \cdot 0.008}{0.95 \cdot 0.008 + 0.05 \cdot 0.992} \approx 0.156 \quad (3.4)$$

Conclusion: Despite a positive test, the chance of having HIV is only 15.6%.

Two Positive Tests

Assuming independence:

$$\begin{aligned} P(\text{HIV}|\text{T}_1, \text{T}_2) &= \frac{P(\text{HIV})P(\text{T}_1|\text{HIV})P(\text{T}_2|\text{HIV})}{P(\text{T}_1)P(\text{T}_2)} \\ &= \frac{0.008 \cdot 0.95 \cdot 0.95}{(0.008 \cdot 0.95 + 0.992 \cdot 0.05)^2} \approx 0.744 \end{aligned} \quad (3.5)$$

Conclusion: Two positive tests increase the probability to 74.4%.

3.8 Types of Naive Bayes Classifiers

- **Gaussian Naive Bayes:** - For continuous data. - Assumes features follow a Gaussian distribution:

$$P(x_i|C) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) \quad (3.6)$$

- **Multinomial Naive Bayes:** - For count data (e.g., word counts). - Used in text classification.
- **Bernoulli Naive Bayes:** - For binary data (e.g., presence/absence of a word).

3.9 Advantages and Limitations

3.9.1 Advantages

- Simple, efficient, and easy to implement.

- Works well on small data and high dimensions.
- Handles categorical and numerical data.

3.9.2 Limitations

- Assumes independence of features.
- Performs poorly when features are correlated.
- Sensitive to zero-frequency problems which has to be solved using Laplace smoothing.

